
LawChain: A Resource-Efficient Blueprint for Legal Information Retrieval in Low-Resource Jurisdictions

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Abstract

We present LawChain, a resource-efficient framework for legal information retrieval in low-resource jurisdictions, developed through a deployment on Sri Lankan legislative Acts. This setting reflects challenges common to public-sector ML in low-resource settings, including many jurisdictions in the Global South: noisy digitized documents, inconsistent legal text structure, limited evaluation benchmarks, and a vocabulary mismatch between lay queries and statutory language. To address these constraints, we build a layout-aware extraction pipeline and a modular retrieval architecture combining BM25, semantic embeddings, localized query expansion, an agentic refinement loop, and a Neo4j knowledge graph. Using a Gemini-based LLM-as-a-judge evaluation framework, the combined system achieves Precision@5 of 0.8345 and Recall@5 of 0.9357. Beyond this deployment, LawChain offers a reusable design pattern for sustainable, locally grounded retrieval under data and compute constraints. We release the system architecture, a 22,812-node legal knowledge graph, and a human-refined dataset of more than 26,000 question-answer pairs to support future work on legal information access in low-resource settings.

1. Introduction

Legal information retrieval systems are critical for enabling equitable access to statutes, case law, and regulatory documents, particularly as legal text volumes grow (Sugathadasa et al., 2018). However, the advancement of these AI-driven

systems has predominantly favored resource-rich jurisdictions, leaving a pronounced access disparity in developing countries characterized by low-resource (de Silva, 2026) and low-access (Wijeratne et al., 2019) data landscapes. In these regions, legal professionals and the general public face systemic socio-technical barriers when querying documents: archives are often highly scattered, inconsistently digitized, and saturated with technical jargon. There is an urgent need not just for localized NLP solutions, but for sustainable, resource-efficient development frameworks that allow underrepresented communities to build and deploy effective legal IR systems without relying on massive compute infrastructure.

Our work addresses this critical gap in legal accessibility by building a comprehensive retrieval platform deployed for Sri Lankan legislative Acts. Rather than force-fitting computationally heavy models onto noisy data, our methodology is deliberately tailored for efficiency and real-world deployability. First, we developed contextually accurate, layout-aware extraction techniques to salvage the structural integrity of poorly digitized legal texts. Building upon this clean foundation, we engineered a context-aware, multi-engine retrieval architecture that leverages the complementary strengths of robust lexical baselines and localized semantic models to bridge the vernacular gap for everyday users.

In order to rigorously evaluate our system and combat the severe lack of evaluation benchmarks in low resource jurisdictions, we compiled heavily curated datasets comprising over 26,000 document-wise and section-wise question and answer pairs. We also generated a graph database of legal acts in Sri Lanka to make retrieval context-aware by explicitly mapping interconnections between documents. To ensure sustainable deployment, our solution utilizes a modular microservice architecture that supports low-overhead horizontal scaling by enabling the integration of diverse retrieval engines. It also accommodates vertical scaling to handle increasing user demand, intentionally designed as a resilient, easily adoptable blueprint for other low-resource environments.

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Thus, the contributions of our work are twofold:

- We design and implement a performant Sri Lankan legal information retrieval system, contributing reusable evaluation datasets, including Q–A pairs and a knowledge graph, to support future work in low-resource legal IR.
- We document our end-to-end development process, design lessons, and scalable microservice-based architecture, providing a practical blueprint for developers in low-resource settings, building retrieval systems under data and compute constraints.

2. Related work

While large language models and dense retrieval have advanced legal informatics, recent legal ML breakthroughs still largely favor resource-rich jurisdictions (Joshi et al., 2020). This section reviews foundational retrieval techniques and the practical barriers that motivate efficient, localized architectures for the Global South.

2.1. Bridging the Vernacular and Structural Divide

Legal information retrieval (IR) differs from general IR because statutory text is both linguistically specialized and structurally complex. Statutes rely on rigid drafting conventions, technical terminology, and long-range dependencies such as cross-references (Krasadakis et al., 2024). In developing nations, these properties can widen the justice gap by creating a vocabulary mismatch between layperson queries and statutory language, often requiring retrieval at a granular section level rather than across whole documents (Westermann et al., 2023).

To bridge this divide sustainably, systems need to combine lexical and semantic signals. BM25 (Robertson & Zaragoza, 2009) remains a computationally efficient and effective baseline, since legal queries often depend on exact phrasing and specific section identifiers (Rosa et al., 2021). Dense retrieval can improve recall when everyday terminology differs from legal phrasing, while hybrid retrieval often outperforms either approach alone (Rayo Mosquera et al., 2025). Large Language Models (LLMs) have also improved legal search interfaces through retrieval-augmented generation (RAG) (Lewis et al., 2020) and LLM-assisted retrieval. Techniques such as HyDE-style query expansion and LLM-based result aggregation (Gao et al., 2023; Zheng et al., 2024) are useful in low-resource settings because they can map informal or vague user queries to formal legal concepts without requiring expensive fine-tuning pipelines.

2.2. Overcoming Low-Resource Infrastructure Barriers

Low-resource jurisdictions face infrastructure constraints that can degrade end-to-end retrieval quality. A central issue

is document digitization: legal sources are often distributed as heterogeneous scanned PDFs. In data-scarce environments, OCR noise and layout artifacts can break section boundaries, degrading both lexical indexing and dense embedding quality (Aumiller et al., 2021; Koreeda & Manning, 2021; Zadgaonkar & Agrawal, 2021).

Relying on off-the-shelf AI tooling is also difficult. Pre-trained legal models often reflect the drafting styles and linguistic patterns of Western jurisdictions, which may not transfer reliably to underrepresented legal corpora (Butler et al., 2025; Chalkidis et al., 2020). Evaluation is similarly constrained by the lack of human-annotated benchmarks. Together, these challenges suggest that successful ML deployments in developing regions should not simply scale up compute. Instead, they should prioritize resource-efficient, structure-preserving extraction, localized hybrid retrieval, and jurisdiction-specific validation to support sustainable and equitable access (Krasadakis et al., 2024).

3. Methodology

To address the resource constraints of the Sri Lankan legal data landscape, we propose a modular, end-to-end retrieval architecture designed for sustainability and low-overhead deployment. The following subsections describe our layout-aware data extraction, hybrid multi-engine retrieval framework, and localized query expansion techniques.

3.1. Data Collection and Preprocessing

Raw legal documentation in Sri Lanka is distributed across multiple publication channels. Senaratna (Senaratna, 2026) aggregated a collection of these documents into a single repository. While Sinhala and Tamil are the national languages (de Silva, 2026), English is used as a link language (Parliament of Democratic Socialist Republic of Sri Lanka, 2022). In this work, we focused on English as the base language while keeping the pipeline language-agnostic for future multilingual incorporation.

Legal texts depend on structure, including section hierarchies and cross-references. Standard PDF extraction methods often fail to preserve this structure, producing fragmented outputs that affect downstream processing.

Rather than relying on compute-heavy OCR correction models, we developed a resource-efficient document extraction pipeline (Figure 1) using a layout-aware parsing approach to access the document’s underlying coordinate system. We applied pattern matching to introductory pages to isolate key metadata, including the Act number, certification year, and preamble. To retrieve the core legal content, we filtered out page numbers and marginal notes based on their horizontal positioning. This content was segmented into discrete sections by identifying standard alphanumeric delimiters.

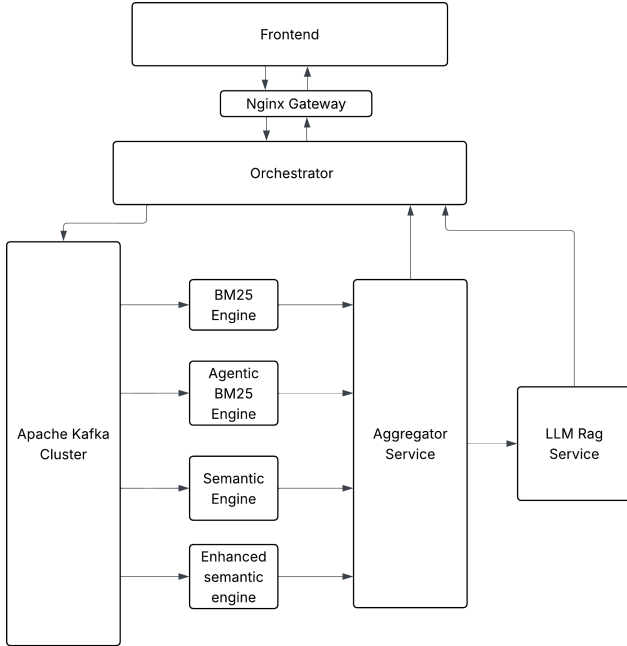


Figure 1. LawChain architecture and retrieval flow.

Finally, we normalized the text to remove layout-induced line breaks, producing a structured text stream for embedding.

3.2. Information Retrieval Architecture

Our approach combines keyword-based retrieval with BM25 (Robertson & Zaragoza, 2009) and semantic retrieval using vector embeddings (Sugathadasa et al., 2018), as shown in Figure 2. BM25 provides a computationally efficient lexical baseline for queries that depend on exact terms (Dhakal et al., 2026), while semantic search supports exploratory queries where users may not know the precise legal terminology (Rayo Mosquera et al., 2025).

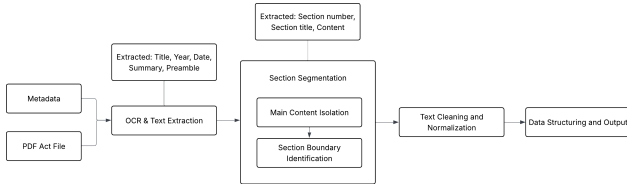


Figure 2. Structure-preserving extraction and section segmentation pipeline for Sri Lankan Acts.

Embedding Model Selection: In the Sri Lankan context, pre-trained models such as Legal-BERT (Chalkidis et al., 2020) did not transfer reliably due to differences in drafting style and document structure (Butler et al., 2025). To balance performance and deployment cost, we evaluated multiple embedding models against our internally constructed Q-A dataset. We selected the **E5-base** model (Wang et al.,

2024) because it provided reliable semantic representations of local legal text while keeping storage and compute requirements low.

Query Expansion: To address the vocabulary mismatch in layperson queries (Westermann et al., 2023), we implemented query expansion (Azad & Deepak, 2019). For BM25, an agentic LLM loop augments queries with statutory terms. If a specialized agent inspects the initially retrieved chunks and deems them irrelevant, it generates a revised set of expansion keywords and re-issues the search (Hayashi et al., 2025; Wang et al., 2023). For the semantic component, we applied Hypothetical Document Embeddings (HyDE) (Gao et al., 2023). We generated hypothetical documents using an instruction fine-tuned LLM trained on our localized Sri Lankan corpus, providing a more grounded semantic signal without requiring exhaustive fine-tuning.

3.3. Knowledge Graph and Aggregation

Legal corpora contain explicit relationships between documents (Krasadakis et al., 2024). We constructed a Neo4j-based knowledge graph to model these links and help users navigate related statutory provisions. We implemented a pipeline that iterated through all documents and identified references such as amendments, repeals, and additions. These extracted links were represented as a graph where edges capture the relationship type, adding broader legal context to retrieval.

To aggregate results from our multi-engine setup, we first experimented with reciprocal rank fusion (RRF) (Cormack et al., 2009), but observed uneven scoring influence. A cross-encoder re-ranking stage improved quality but was too memory-intensive and slow for low-resource environments. Finally, we adopted an LLM-as-a-judge (Zheng et al., 2024) using Gemini 2.0. This method outperformed previous iterations by providing more consistent cross-engine ranking while avoiding the overhead of a cross-encoder.

3.4. Retrieval Augmented Generation (RAG) & Application Modularity

To improve accessibility for users without legal training, we adopted a RAG approach in which the final ranked passages are provided to Gemini 2.0 to generate a clear, compact summary grounded in the retrieved source material.

To support long-term deployability, the platform was built using a microservices architecture. Inter-service communication is handled through an Apache Kafka message broker, enabling reliable message passing. Individual retrieval engines operate independently, allowing researchers in developing nations to substitute embedding models or deploy the system across distributed, low-cost hardware rather than relying on monolithic servers. This makes the system an

extensible and scalable blueprint for similar low-resource settings.

4. Discussion and Conclusion

Assessing ML systems in developing regions requires attention to both empirical accuracy and real-world use. This section outlines our evaluation framework, analyzes the role of our extraction pipeline, and discusses how the architecture can serve as a scalable blueprint for similar jurisdictions.

4.1. Datasets & Experimental Setup

To address the lack of evaluation benchmarks in low-resource jurisdictions, we generated three datasets:

- **Graph dataset:** A corpus with 22,812 section nodes and 300+ connections capturing inter-document links.
- **Section-wise Q-A dataset:** 1,600 pairs where the ground-truth answer is a specific section.
- **Act-wise Q-A dataset:** 25,000+ pairs acting as our “silver dataset.” Questions were generated using an LLM based on the original legal acts and subsequently refined by human reviewers to ensure relevance.

Evaluation Methodology: To evaluate system performance without relying on expensive, large-scale human annotation, we used Gemini 2.0 as an LLM-as-a-judge. Recall was calculated based on the proportion of queries for which the system retrieved the correct act from the dataset. For precision, if the exact act was not directly retrieved, the retrieved chunk was passed to the LLM to assess its semantic relevance to the query. Finally, success rate was defined as the proportion of queries where at least one retrieved chunk was deemed highly relevant.

4.2. Results & Discussion

We evaluated Precision@5, Recall@5, and Success@5 for each retrieval engine (Table 1). The integrated pipeline outperforms the individual engines, suggesting that combining lexical baselines with localized semantic retrieval is useful in low-resource environments.

Table 1. Retrieval performance at top-5 (per-engine) with combined system results.

Engine/System	P@5	R@5	Success@5
BM25 Engine	58.4%	67.1%	89.2%
Semantic Engine	71.8%	82.6%	92.8%
Enhanced Semantic	74.2%	85.9%	94.1%
Agentic BM25	69.7%	78.3%	92.6%
Combined System	83.45%	93.57%	95.58%

A key contributor to this performance is the layout-aware extraction pipeline. By embedding coherent sections rather than fragmented pages, the representation space separates recurring legal boilerplate from statute-specific content.



Figure 3. 2D projection of embedded sections from Sri Lankan Acts, colored by Act.

As shown in Figure 3, structural provisions that recur across Acts cluster together, reducing the extent to which boilerplate dominates similarity computations and contributes to retrieval noise.

4.3. Conclusion

In this work, we demonstrated the feasibility of building a strong legal IR system for a low-resource jurisdiction. By treating unstructured digitization and vocabulary mismatch as design constraints rather than only downstream engineering issues, we developed a pipeline optimized for sustainable deployment. LawChain provides an end-to-end, scalable reference design for improving legal information access across similar low-resource settings.

5. Future work

Future work will focus on three directions. First, we plan to integrate low-compute OCR methods to recover degraded archival scans that are not handled by the current extraction pipeline. Second, we will extend the system to support cross-lingual retrieval and generation in Sinhala and Tamil, improving access for non-English-speaking users. Finally, we will add stronger citation-grounded guardrails in the RAG module to reduce hallucinations and improve reliability in legal settings.

Impact Statement

This work aims to improve access to legal information in low-resource jurisdictions by developing retrieval systems that can be deployed and evaluated under local constraints. Such systems may benefit legal practitioners, researchers and members of the public. LawChain is intended to support legal information access, not replace professional legal advice.

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